

ACE6313 Machine Learning Concepts and Technologies

Trimester March/April 2026 (T2610)

Assignment (40%)

Objectives

- Apply the knowledge learned from the course
- Acquire experience in developing machine learning applications
- Build machine learning applications to solve real-world problems
- Appreciate the importance of coordinated teamwork and project management

Summary

- Date released: Week 4
- Create Group: Week 5-6
- Date of interim assessment: Week 7-8
- Date of submission and presentation: Week 14
- Assessment marks: 40%

Introduction

In this assignment, each group will explore how machine learning can support one of the United Nations' Sustainable Development Goals (SDGs). The goal is to apply machine learning techniques to analyze and derive meaningful insights from real-world data, while gaining hands-on experience in data preprocessing, model building, and hyperparameter tuning.

Getting Started

1. Group Composition

- Each group consists of 2 students.
- One student will focus on data preprocessing.
- The other student will focus on machine learning models.

2. Topic Selection

- Propose a topic that aligns with any one of the UN's SDGs (<https://sdgs.un.org/goals>).
- Example: Reducing food waste (SDG 12), predicting water quality (SDG 6), or analyzing urban mobility (SDG 11).
- Each group must have different topics.

3. Dataset

- Use a real-world dataset from sources such as Kaggle, UCI Machine Learning Repository, or public datasets relevant to your topic.
- Each group must use a unique dataset, to be finalised on a first-come, first-served basis.

Requirements:

Part A: Data Preprocessing

The student for this part must develop a data preprocessing pipeline comprising the following stages:

1. Data Cleaning
 - Handle missing values, duplicates, inconsistencies and/or outliers.
2. Data Transformation
 - Perform feature scaling, encoding and/or feature engineering.
3. Data Reduction
 - Apply techniques like feature selection, feature extraction, or dimensionality reduction (e.g., PCA).

Part B: Machine Learning Models

The student for this part must:

1. Train and evaluate at least five different machine learning algorithms, such as:
 - Linear Regression/Logistic Regression
 - Decision Trees
 - Random Forest
 - Support Vector Machines (SVM)
 - k-Nearest Neighbors (kNN)
 - Gradient Boosting Trees
 - Neural Networks
2. Apply hyperparameter tuning for model optimization.
3. Compare the performance of the models using appropriate metrics (e.g., accuracy, precision, recall, F1-score, R^2 score or RMSE).

Part C: Collaboration and Documentation

1. Use GitHub for collaboration where each student must have meaningful contributions documented in the repository.
2. Write a detailed final report (6 to 10 pages) covering:
 - Introduction.
 - Data preprocessing details.
 - Machine learning models.
 - Results discussion.
 - Conclusion.

Mode of submission

- Interim assessment (Week 7-8):
 - Finalized proposal
 - GitHub repository
 - Dataset description
- One final report (softcopy) per group (please follow the following format):
 - A4-size pages (single spacing with 12-pt Times New Roman; **min** 6 pages, **max** 12 pages excluding cover, source codes and appendices).
 - The cover should show the **subject code and name, name, student ID and email** of each member.
 - **ONE PAGE** per group member, individually authored, outlining that person's reflections on what he/she has contributed to the team effort and what he/she has gained from his/her participation (must be **SPECIFIC**).
- GitHub repository to show the progress (commits, documentation, logs of changes and updates) chronologically, with all Python scripts or Jupyter Notebooks and data files (links if data files are too large).
- One zip file containing the following items:
 - i. Assignment report in Microsoft Word (docx) format.
 - ii. Presentation slides in PowerPoint (pptx) format.
- Name the zip file using your group number, e.g. Group 1.zip
- File submission via uploading to eBwise.
- Due date: Monday of Week 14.
- Presentation: Week 14
- Late submission will be penalised accordingly.

Assessment Scheme (Total 40%)

Report (20%)

An introduction with background, objectives, problem statement and deliverables
Explanation of the data preprocessing (dataset description, pipeline stages)
Explanation of machine learning models (training and evaluation)
Discussion of results (insights derived from the analysis)
A conclusion summarizing the achievement, limitations and future work
One page on personal reflections (one page per member)

Project Progress (5%)

Evidence of progress (GitHub commits frequency, logs of updates and changes, documentation)

Presentation (15%)

Data preprocessing
Machine Learning Models
Individual Q & A